Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18ME51]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

MANUFACTURING PROCESS-II

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Draw Diagram wherever necessary.

1.	a.	Define explosive welding. Draw a schematic diagram and note down the process parameters.	10
	b.	Explain the principle of generation of LASER.	04
2.	a.	Describe the principle and equipment for Water Jet Machining. Enlist its applications and process parameters.	10
	b.	Compare conventional and Non-conventional machining process.	04
3.	a.	Define Electro Chemical Machining Process. Draw a schematic diagram and list the process parameters.	10
	b.	List the advantages and limitations of EDM process.	04
		OR	
4.	a.	Briefly explain the following with respect to chemical machining: i. Characteristics of cut peel maskants ii. Selection of maskants iii. Advantages of photoresist maskant iv. Limitations of chemical machining.	10
	b.	Compare CHM and ECM.	04
5.	a.	Discuss the importance of surface hardening processes by taking an example of a machine component.	10
	b.	Compare PVD & CVD process.	04
		OR	
6.	a.	With a neat sketch, explain electroplating process. Also mention advantages, disadvantages and applications.	10
	b.	Discuss the advantages and limitations of Anodizing process.	04
7.	a.	Illustrate investment casting process with a neat sketch. Also mention advantages, disadvantages and applications.	10
	b.	Explain the various pattern allowances.	04
		OR	
8.	a.	Enlist and explain the materials used in pattern making. Also explain any four types of pattern in detail.	08
	b.	With a neat diagram, explain the process of shell moulding.	06